Lab Report: Finding the Optimal Price

Problem Summary

The purpose of this assignment was to identify the optimal price that maximized expected revenue per customer while minimizing the number of customers used in testing. The optimal price existed somewhere between $0 and $70. However, no information existed on how price affected customer demand. In order to identify this relationship, I had to run experiments at different prices to determine which price would generate the most amount of revenue. One challenge was that multiple prices could appear optimal, even though only one was the global optimum. The goal was to find a balance between exploring different prices and generating as much revenue as possible.

Experimental Design

I ran an initial grid search from $0 through $70, at intervals of $5, testing approximately 500 customers per price level. I identified potential price levels and performed a more focused search from $20 through $35 at $1 increments. I also increased the sample size for selected prices to obtain more reliable revenue estimates. Since the demand function could have multiple local maximums, I explored additional prices outside the initial range to help me identify another high performing region around $12. Throughout the experiments, I tracked the number of customers tested and calculated average revenue per customer.

After analyzing the results of my initial experiments, I developed a more efficient adaptive approach. I started with smaller samples and increased the sample size for prices that showed high performing results instead of using large samples of customers for every price. I then improved this approach through an automated pricing method that selected prices based on previous experimental results. The algorithm performed a broad grid search using approximately 100 customers per price and automatically identified three high performing price regions. It then

tested nearby prices within each region using approximately 150 customers per price. Then it selected the five best performing prices overall and retested them using approximately 400 customers each to obtain more reliable estimates. I then combined the results from previous experiments with new observations to make better use of the data already collected.

Challenges and Improvements

One of my biggest challenges was the unpredictability in customer purchases. While a certain price may have generated significant sales during an experiment, it would produce significantly different results in another test. For example, prices around $26–$28 were consistently high performing, but the best observed price changed between experiments, making it difficult to determine whether a price was actually better or whether its performance was influenced by random variation. As a result, I increased sample sizes for the high performing prices and combined observations from repeated tests, which allowed me to compare prices using more information.

An additional problem in determining the best pricing strategy was balancing the number of customers tested with the revenue generated. My initial approach allocated customers to less profitable prices, which reduced revenue efficiency. I then changed my approach to use smaller samples to see where I should look next to then focus additional tests on those areas. I also wanted the algorithm to be automated for reproducibility. Using automation allowed the final version of the algorithm to systematically test various price ranges while keeping track of the cost associated with the testing.

Results and Conclusion

price	customers	revenue	avg_revenue
28	1184	14392	12.16
26	652	7644	11.72
29	1131	13108	11.59
25	837	9600	11.47
12	1060	12096	11.41

Table 1

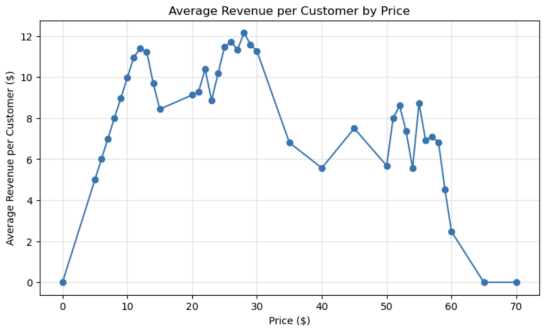


Figure 1

Table 1 shows the five prices that generated the highest average revenue per customer during my final automated experiment. Figure 1 shows how average revenue varied across the prices tested. The automated pricing algorithm identified $28 as the best performing price, generating approximately $12.16 per customer. In addition to testing 13,659 potential customers, the automated pricing algorithm generated $129,748 in experimental revenue, resulting in an overall average revenue of approximately $9.50 per customer. Since the theoretical optimal expected revenue is $11.925 per customer, this represents about 79.66% of optimal revenue. The manually created price selection strategy achieved a higher percentage, however, it relied on data collected from previous experiments and manual price selection. The fully automated strategy was my final solution because it systematically selected prices without requiring manual intervention.